Aymen Qabel

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Education

Columbia University New York, USA
Ph.D. in Computer Science Sep 2023 – Present
- Deep Generative Models, Machine Learning Theory, Algorithms.
- **Teaching:** Machine Learning for Functional Genomics (TA).

École Normale Supérieure (ENS) / Univ. Paris-Saclay Gif-sur-Yvette, France
Master of Science, MVA Sep 2021 – Sep 2022
- Probabilistic Modeling; Convex Optimization; Graph learning; NLP/Speech Recognition, Kernel Methods

CentraleSupélec / Univ. Paris-Saclay Gif-sur-Yvette, France
Diplôme d'Ingénieur, Data Science Sep 2018 – Sep 2021
- ML/Optimization, Time Series, Graphical Models, Stochastic Processes, Advanced Probabilities, NLP, Computer Vision, Big Data
- Exchange semester at UT Austin (probability/PDEs/data science).

Research & Industry Experience

Graduate Researcher - AQLab, Columbia University | NYC, USA Sep 2023 – Present
- Co-developed **OpenFold3**† (open-source AlphaFold3 reproduction) for joint 3D modeling of proteins, nucleic acids, ligands, and complexes.
- Developed diffusion/flow-matching methods and evaluation with emphasis on stability, calibration, and reproducibility.

Quantitative Researcher - Optiver | Chicago, USA Jun 2026 – August 2026
- Developed strategies for low latency execution of orders and their success.

Research Engineer - DaSciM, LIX, École Polytechnique | Palaiseau, France May 2022 – Aug 2023
- Structure-aware antibiotic resistance classification using **Graph Neural Networks**; PyTorch Geometric pipelines and robust validation.

Machine Learning Research Intern - Amadeus | Biot, France Apr 2021 – Sep 2021
- Demand/booking forecasting under non-stationarity (COVID): features, evaluation design, monitoring.

NLP Research Intern - Institut Gustave Roussy | Villejuif, France Oct 2020 – Apr 2021
- Developed **KmemBERT** for explainable survival prediction from oncology reports.

Publications

1. **Confornets: Latents-based conformational control in openfold3**. *preprint*.
2. **OpenFold3**†. Research preview / software release, 2025. *GitHub — Zenodo*.
3. **Structure-Aware Antibiotic Resistance Classification Using Graph Neural Networks**. *NeurIPS 2022 AI4Science Workshop*. *OpenReview*.
4. **An explainable language model predicts survival from medical reports in oncology**. *Preprint*. **Co-first author**. *SSRN*.

Skills

Programming: Python (NumPy, SciPy, Pandas, scikit-learn, PyTorch, PyTorch Geometric), SQL, Scala, R, MATLAB; JAX (basic).
Tools: Linux, Git, Docker, Slurm/HPC, experiment tracking, reproducible pipelines.
Languages: Arabic (native), French (bilingual), English (fluent).

Awards & Leadership

- First in Moroccan Scientific Baccalaureate.
- Top 2% nationally (Concours National Commun 2018, Morocco).
- French Government Major–Excellence Scholarship.
- Co-organizer, International Research & Innovation Seminar (IRIS), UM6P Morocco (Dec 2023 & Jun 2025).